

SMART INDIA HACKATHON 2026 · PROBLEM STATEMENT 26038

Explainable AI for Diabetic Retinopathy Screening in Rural India

Technical Design Document

Organisation: MathWorks · **Category:** Software

Version: 1.0 · **Date:** 21 August 2026

Status: Baseline for team execution

Presenting member: Rithviksai

0. How to Use This Document

This is the team's single source of truth. Every technical decision recorded here carries the reason it was made, because at judging you will be asked to defend choices, not to prove you typed the code.

0.1 Reading paths

If you are	Read
Presenting (Rithviksai)	§1, §2, §3, §11, §17, and the "Design rationale" subsection of §5–§9. These contain every answer to a likely judge question.
Building the ML pipeline	§5–§8, §10, §11, §13, §14
Building the Simulink model	§9, §10.1, §12.3
Writing the pitch deck	§1, §2.3, §11.1, §16, §17

0.2 Evidence marking convention

The team asked explicitly for content that is real and applicable rather than plausible-sounding. Every factual claim in this document is therefore tagged:

Tag	Meaning
VERIFIED	Confirmed against a primary source during document preparation. Source listed in §18.
ESTABLISHED	Standard, widely reproduced knowledge in the field (e.g. the ICDR grading rules). Safe to state, but cite the primary reference in §18 before putting a number on a slide.
UNVERIFIED	Believed correct but not confirmed. Do not put these on a slide or in the SIH submission until a team member has checked the primary source. Tracked in Appendix B.

THE ONE RULE THAT MATTERS MOST

If a number is tagged Unverified and nobody has checked it by the time you present, **delete it**. A missing statistic costs you nothing. A fabricated one, challenged by a judge who knows the literature, ends your run. This applies with equal force to model results: report what you measured, not what you hoped for.

0.3 Document control

Owner	Team lead
Revision trigger	Any change to the validation protocol, the frozen operating point, or the dataset split
Frozen sections	§10.4 (sealed test set) and §11.2 (operating point) must not be edited after the freeze date without recording the change and the reason in this table

1. Executive Summary

1.1 What we are building

A MATLAB-based retinal image analysis pipeline that screens fundus photographs for diabetic retinopathy at primary health centres in rural India, and that explains every decision it makes in terms a clinician can check in under thirty seconds.

The pipeline has five stages, matching the five requirements of SIH26038 one-to-one: an image quality gate that rejects ungradable captures and tells the operator how to retake them; a segmentation stage that finds the optic disc, fovea, vessels and the four lesion classes that define DR severity; a grading stage that outputs an International Clinical Diabetic Retinopathy (ICDR) severity level with a calibrated confidence; an explainability module that produces Grad-CAM attention maps *and* lesion-level clinical evidence *and* an annotated referral report; and a SimEvents discrete-event model of the district telemedicine programme that sizes grader and bandwidth capacity for a population of 100,000+ screened per year.

1.2 The distinctive claim

The problem statement says existing AI solutions "function as black boxes, lack clinical validation rigor, and fail with variable image quality from portable fundus cameras in field conditions." Those are three separate charges, and our design answers each with a concrete mechanism rather than a claim.

PS charge	Our mechanism
"function as black boxes"	Two independent explanation channels, and a measured comparison between them. Grad-CAM (post-hoc, mandated by the PS) is benchmarked against inherently interpretable lesion-level evidence on the same images, scored quantitatively against IDRiD pixel-level lesion masks. Where the two disagree, the case is flagged as an unexplained prediction and routed to a human. Explanation stops being decoration and becomes a safety control.
"lack clinical validation rigor"	Messidor-2 is sealed before development begins and opened exactly once, after the operating point has been frozen on a separate validation split. Results are reported at a pre-committed threshold with confidence intervals, per-class recall, and calibration error — never as bare accuracy. If external performance falls short of the PS targets, we report and analyse that rather than tuning until it looks good.
"fail with variable image quality"	A quality gate runs <i>before</i> the classifier is permitted to predict, and can refuse. Every downstream metric is additionally reported stratified by quality tier, so the reader can see how performance degrades as capture conditions worsen instead of seeing a single number averaged over a clean subset.

1.3 Why the explanation channel is worth building

The value of explanations in DR screening is not an assumption we are making. It has been measured. A 2025 study in *PLOS Digital Health* compared an inherently interpretable model (a sparse BagNet, whose receptive field is deliberately capped so that its evidence map *is* its reasoning) against post-hoc saliency, and ran a reader study with six ophthalmologists. **VERIFIED**

Measurement	Result	Why it matters to us
Lesion localisation precision, sparse BagNet	0.960	Post-hoc saliency is substantially worse at landing on the actual lesion. This is the evidence base for building a second, lesion-grounded channel rather than shipping Grad-CAM alone.
Lesion localisation precision, Guided Backpropagation	0.656	

Measurement	Result	Why it matters to us
Ophthalmologist accuracy, unaided	0.611	Explanations add accuracy <i>on top of</i> the prediction (0.758 → 0.786). The AI does not merely automate; the explanation itself carries information the grader uses.
... with AI prediction only	0.758	
... with AI prediction and explanation	0.786	
Time per case with explanations	11.7 s vs 15.2 s	~25% faster. Maps directly onto the PS requirement of "ophthalmologist validation in under 30 seconds."

1.4 What most competing teams will not build

Requirement 5 of the PS — the Simulink workflow simulation — is a fifth of the problem statement and is almost universally skipped or reduced to a static bar chart. It requires no GPU, no dataset and no machine learning. It is the single cheapest differentiator available, and it is the MathWorks-specific part of a MathWorks problem statement.

It is also genuinely interesting, because it closes a loop nothing else in the pipeline closes: the classifier's operating point is an *input* to the capacity model. Raising sensitivity produces more false positives, which consumes more grader hours, which lengthens turnaround. There is an optimum, and we can find it and plot it. See §9.

2. Problem Statement Decomposition

2.1 Source text

Problem Statement ID	26038
Title	Explainable AI for Diabetic Retinopathy Screening in Rural India
Organisation / Department	MathWorks
Category	Software
Theme (as listed on portal)	Clean & Green Technology
Named datasets	APTOS 2019 Blindness Detection; IDRiD; DRIVE; Messidor-2
Named toolboxes	Image Processing, Computer Vision, Deep Learning, Medical Imaging, Simulink, Statistics and Machine Learning

NOTE ON THE THEME TAG

The portal lists this MedTech problem under *Clean & Green Technology*. This appears to be a portal categorisation artefact rather than a substantive instruction. It matters operationally because it determines which theme bucket you are judged in and how many teams you compete against. **Action:** confirm the judging bucket with your SPOC before 30 August. Do not attempt to re-frame the project as an environmental one; the PS body is unambiguously clinical.

2.2 Requirements traceability

Every requirement below is extracted verbatim in substance from the PS. The right-hand column is the section of this document that discharges it. Nothing in the PS is unaddressed; where we cannot fully deliver a requirement with the available data, that is stated explicitly rather than glossed (see R2.6).

ID	Requirement (from PS)	Type	Section	Risk
R1.1	Automatically evaluate fundus images for adequacy: focus, illumination, field of view	Functional	§5.2	Low
R1.2	Apply adaptive enhancement (CLAHE, illumination normalisation, denoising) for borderline images	Functional	§5.4	Low
R1.3	Reject ungradable images with recapture feedback	Functional	§5.5	Low
R2.1	Optic disc and fovea localisation	Functional	§6.2	Low
R2.2	Vessel segmentation	Functional	§6.3	Low
R2.3	Microaneurysm detection ("sub-pixel")	Functional	§6.4	High
R2.4	Exudate segmentation	Functional	§6.5	Medium
R2.5	Haemorrhage classification	Functional	§6.5	Medium
R2.6	Neovascularisation detection	Functional	§6.6	High — data gap, see §17.1
R3.1	Classify on ICDR scale, Levels 0–4	Functional	§7.3	Low
R3.2	>90% sensitivity for referable DR (Level 2+)	Performance	§11.2	High
R3.3	>85% specificity for referable DR	Performance	§11.2	High
R4.1	Grad-CAM attention maps	Functional	§8.2	Low

ID	Requirement (from PS)	Type	Section	Risk
R4.2	Lesion-level evidence correlated with clinical criteria	Functional	§8.4	Medium
R4.3	Calibrated confidence scores	Functional	§7.6	Low
R4.4	Automated annotated reports	Functional	§8.7	Low
R4.5	Ophthalmologist validation in under 30 seconds, human-in-the-loop	Performance	§12	Medium
R5.1	Simulink model of telemedicine pipeline: acquisition rates, bandwidth, throughput, review capacity	Functional	§9.3	Low
R5.2	Optimise resource allocation for district programmes serving 100,000+ patients annually	Functional	§9.5	Low
R6.1	Validation against published benchmarks	Evidence	§11.5	Medium
R6.2	Demonstrate the integrated pipeline outperforms any single-technique approach	Evidence	§11.6	Medium
R6.3	Grad-CAM outputs rated as clinically useful	Evidence	§12	Medium

2.3 Reading R6.2 correctly

R6.2 is the requirement teams misread. "Outperforms any single-technique approach" is not rhetorical padding — it is an instruction to run an **ablation study**. You are being asked to build the components separately, measure them separately, then measure them combined, and show the combination wins.

This has a direct architectural consequence and it is why §11.6 exists. We must be able to evaluate, on the same held-out data:

1. Classifier alone, no quality gate, no lesion evidence.
2. Classifier with quality gate.
3. Lesion-rule grading alone (ICDR criteria applied to detected lesion counts, no CNN).
4. Full integrated pipeline: quality gate + classifier + lesion evidence + deferral.

Configuration 3 will almost certainly be the weakest on raw accuracy, and that is fine — its purpose is to be the interpretable reference, and its disagreements with configuration 1 are what drive the unexplained-prediction flag. The pipeline must therefore be built so that any stage can be switched off from a config file rather than by editing code. See §13.3.

2.4 Statistics quoted in the problem statement

The PS supplies its own framing statistics. Use these as primary in the pitch, because quoting the PS back demonstrates you read it. One supporting figure is worth adding to show you read past it.

Statistic	Source	Use
India has over 77 million diabetic adults, second highest globally	PS text	Primary framing VERIFIED as PS content
DR affects ~18% of this population	PS text	Primary framing VERIFIED as PS content
Early screening can prevent 90% of vision loss	PS text	Primary framing VERIFIED as PS content
~1 ophthalmologist per 100,000 rural population	PS text	Primary framing. This is the number that motivates §9's capacity model. VERIFIED as PS content
~101 million adults with diabetes in India (ICMR-INDIAB)	<i>Lancet Diabetes Endocrinol</i> , 2023	Supporting. Shows the PS's 77M figure is the older IDF estimate and the burden is larger still. UNVERIFIED — check before use

Framing note for the pitch. The PS's own numbers already make the argument: roughly 77 million diabetic adults, ~18% with DR, and one ophthalmologist per 100,000 rural people. You do not need to embellish. Present the arithmetic, not adjectives.

3. Clinical Foundations

You cannot defend a grading model without knowing the grading rules it is imitating. This section is the minimum clinical literacy every team member needs, and it is where most of the judge questions will come from.

3.1 What diabetic retinopathy is

Chronic hyperglycaemia damages the small blood vessels of the retina. The vessel walls weaken and leak, then occlude, starving areas of retina of oxygen. In response the retina releases growth factors that drive the formation of new, fragile vessels. Those vessels bleed into the vitreous and pull on the retina as they contract, which is the mechanism of severe vision loss.

The clinically important consequence for us: **DR is graded by counting and locating lesions, not by looking at the whole image holistically.** That is why a lesion-grounded explanation is not merely a nice presentation choice — it mirrors the actual diagnostic procedure a human grader performs. **ESTABLISHED**

3.2 The lesions, in the order they appear

Lesion	Abbr.	Appearance	Typical size	Significance
Microaneurysm	MA	Small round red dot	~10–100 µm	The earliest visible sign. MAs alone define Mild NPDR. This is the hardest object in the whole problem and it drives our entire resolution strategy.
Dot / blot haemorrhage	HE	Red, irregular, larger than MA	~100–500 µm	Counting haemorrhages per quadrant is central to the severe-NPDR rule.
Hard exudate	EX	Yellow, sharp-edged, waxy	Variable	Lipid deposits from leaking vessels. Near the fovea, implies macular oedema.
Soft exudate / cotton-wool spot	SE	Pale, fluffy, indistinct edges	Variable	Nerve-fibre-layer infarcts. Marker of ischaemia.
Venous beading	VB	Sausage-like vein calibre variation	Vessel-scale	Part of the severe-NPDR rule. Requires vessel segmentation to assess.
Intraretinal microvascular abnormality	IRMA	Tortuous intraretinal vessel loops	Vessel-scale	Part of the severe-NPDR rule. Very hard to distinguish from neovascularisation.
Neovascularisation	NV	Fine new vessel networks, often at the disc	Vessel-scale	Defines Proliferative DR. Sight-threatening. No pixel-level ground truth exists in our named datasets — see §17.1.

THE MICROANEURYSM RESOLUTION PROBLEM — READ THIS BEFORE CHOOSING ANY INPUT SIZE

A microaneurysm can be as small as $\sim 10\ \mu\text{m}$. In a fundus photograph, that is on the order of a handful of pixels at native camera resolution, and it becomes *sub-pixel* at the input sizes CNNs conventionally use.

Consider what actually happens with a standard pipeline. An IDRiD image is 4288×2848 . Resize it to the ResNet-50 default of 224×224 and you have shrunk it by a factor of roughly 19 in each dimension. A 20-pixel microaneurysm becomes approximately one pixel. It is not that the model finds this hard — **the evidence has been destroyed before the model ever sees it.**

This single fact drives three decisions later in this document: the classifier operates at high input resolution (§7.2), lesion segmentation is patch-based at native resolution rather than on downsampled whole images (§6.4), and Grad-CAM's coarse output is explicitly acknowledged as insufficient for MA localisation rather than being oversold (§8.3).

3.3 The ICDR severity scale

The PS mandates the International Clinical Diabetic Retinopathy severity scale, Levels 0–4. These are the rules our Level 0–4 output must correspond to. **ESTABLISHED**

Level	Name	Definition	Referable?
0	No apparent retinopathy	No abnormalities	No
1	Mild NPDR	Microaneurysms only	No
2	Moderate NPDR	More than microaneurysms only, but less than severe NPDR	Yes
3	Severe NPDR	Any of the 4-2-1 rule below, with no signs of proliferative disease	Yes
4	Proliferative DR	Neovascularisation, and/or vitreous or preretinal haemorrhage	Yes

The 4-2-1 rule (defines Level 3)

Severe NPDR is diagnosed if *any one* of the following is present: **ESTABLISHED**

- **4** — more than 20 intraretinal haemorrhages in each of all four quadrants; *or*
- **2** — definite venous beading in two or more quadrants; *or*
- **1** — prominent IRMA in one or more quadrants.

WHY THE 4-2-1 RULE IS THE BACKBONE OF OUR EXPLAINABILITY MODULE

Notice what the rule is made of: **counts of specific lesions within specific spatial regions**. That is precisely what a lesion segmentation model produces. This means our lesion-evidence channel (§8.4) is not an approximation of clinical reasoning invented for a hackathon — it is a direct computational restatement of the published grading criterion. When a judge asks "how do you know your explanation is clinically meaningful?", the answer is: because it is expressed in the same terms as the diagnostic standard, and here is the rule it evaluates.

3.4 Referable DR: the endpoint that actually matters

The PS sets its performance targets on **referable DR, defined as Level 2 or above**. This matches international screening practice and is the endpoint used by regulator-cleared autonomous systems. The screening question is not "what grade is this?" but "does this person need to see an ophthalmologist?"

This asymmetry must shape every evaluation decision we make:

Error	Consequence	Implication
False negative (missed referable DR)	Patient is told they are fine and does not return for a year. Disease progresses. Potentially irreversible vision loss.	Catastrophic and irreversible. This is why the PS sets the sensitivity bar (>90%) higher than the specificity bar (>85%).
False positive	Patient is referred unnecessarily. One wasted specialist appointment; patient travel and time cost.	Costly but recoverable. Quantifiable — and in §9 we actually quantify it, in grader-hours per year.

NEVER REPORT BARE ACCURACY

In APTOS, roughly half the images are Level 0. A model that predicts "no DR" for every image scores respectably on accuracy while being clinically worthless and dangerous. Any result in this project that is reported as a single accuracy figure is, by team policy, an invalid result. §11 defines what we report instead. This is also the failure mode most likely to slip past an AI-assisted workflow silently, because a model that learns the majority class throws no error — see §13.4.

3.5 Benchmark performance from cleared systems

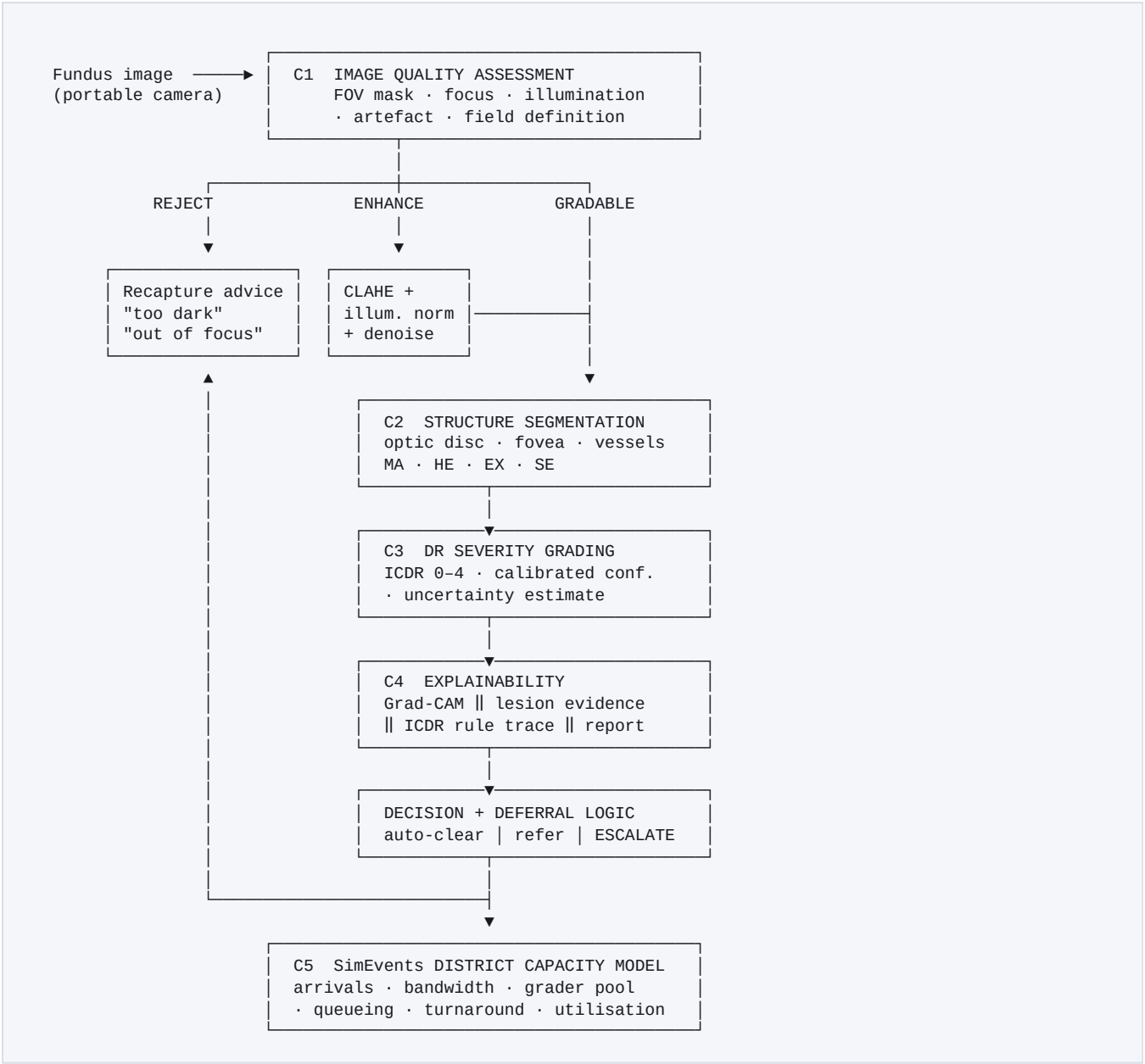
These give context for how hard the PS targets are. Regulator-cleared autonomous DR screening systems provide the realistic comparison. **UNVERIFIED — CONFIRM FIGURES BEFORE CITING**

System	Sensitivity	Specificity	Note
IDx-DR (pivotal trial)	87.2%	90.7%	First FDA De Novo authorised autonomous DR system (2018)
EyeArt (pivotal)	~96%	~88%	FDA 510(k), 2020
SIH26038 target	>90%	>85%	Set by the PS

Read that table carefully, because it contains an important and slightly uncomfortable message. **The PS's sensitivity target exceeds what IDx-DR achieved in its pivotal trial.** These targets are achievable on an internal test split of a single dataset. They are considerably harder on a genuinely external dataset from different cameras and a different population. §11.4 explains how we handle that honestly rather than by quietly reporting only the internal number.

4. System Architecture

4.1 Pipeline overview



4.2 The three-way decision, and why deferral is a first-class output

Most DR classifiers emit two outcomes: refer or do not refer. Ours emits three.

Outcome	Triggered when	Action at the PHC
Auto-clear	Confident Level 0–1, lesion evidence agrees, image quality good	Patient advised of routine annual re-screening. No specialist time consumed.
Refer	Level 2+ predicted, and lesion evidence supports it	Referral slip printed with annotated image and the ICDR criterion that fired.

Outcome	Triggered when	Action at the PHC
Escalate (human grading required)	Model uncertainty above threshold, or Grad-CAM and lesion evidence disagree, or image quality borderline	Image queued for remote human grader. This is the human-in-the-loop path the PS requires under R4.5.

DESIGN RATIONALE: THE "UNEXPLAINED PREDICTION" FLAG

This is the mechanism we consider the most defensible original contribution in the design, and it is worth being able to explain fluently.

If the CNN predicts referable DR but the lesion detector finds no microaneurysms, no haemorrhages and no exudates, then one of two things is true: either the lesion detector missed something, or **the classifier is keying on something that is not disease**. The second case is a well-documented failure mode in retinal imaging — models learn camera artefacts, image borders, and laterality rather than pathology.

Either way, the case is one a human should see. Standard post-hoc XAI would generate a heatmap for this prediction and present it as a justification, which is actively misleading. Our design detects the condition and refuses to auto-decide. **This converts explainability from a presentation layer into a runtime safety control**, and it is measurable: §11.7 defines the disagreement-rate metric.

4.3 Deployment context

The PS specifies deployment "in primary healthcare centres across rural India" with "portable fundus cameras in field conditions." That setting imposes constraints, and each constraint has a concrete design consequence:

Field reality	Design consequence
Operator is a health worker, not an ophthalmologist	Explanations must be readable without specialist training. The referral slip states a plain-language reason and the ICDR criterion, not a bare probability.
Handheld/portable cameras, uncontrolled lighting	Quality gate is mandatory and runs first. Performance reported stratified by quality tier (§11.3).
Intermittent or absent connectivity	Inference must run locally. Model size and inference latency are budgeted, not incidental. Store-and-forward queue when the link is down — modelled explicitly in §9.
Patient may not return	Result must be produced during the visit. This sets the end-to-end latency target that §9 optimises against.
Limited specialist capacity downstream	Every false positive consumes scarce grader time. §9 quantifies this in grader-hours per year and finds the operating point that respects it.

4.4 Toolbox mapping

The PS names six MathWorks products. Depth of toolbox use is a scoring dimension in a MathWorks problem statement, so this mapping is deliberate rather than incidental.

Product	Component	Principal use
Image Processing Toolbox	C1, C2	FOV masking, <code>adapthisteq</code> (CLAHE), morphological lesion candidate detection, denoising, illumination normalisation
Computer Vision Toolbox	C2	Semantic segmentation networks, labelling and evaluation utilities
Deep Learning Toolbox	C2, C3, C4	<code>trainnet</code> , <code>dlnetwork</code> , <code>imagePretrainedNetwork</code> , and the three XAI functions: <code>gradCAM</code> , <code>imageLIME</code> , <code>occlusionSensitivity</code> VERIFIED

Product	Component	Principal use
Medical Imaging Toolbox	C2, C4	Medical image datastores, annotation and visualisation of clinical overlays
Statistics and Machine Learning Toolbox	C1, C3, C4	Quality-tier classifier, probability calibration, confidence intervals, ROC analysis
Simulink + SimEvents	C5	Discrete-event district capacity model. SimEvents supplies Entity Generator, Queue, Server, Resource Pool / Acquirer / Releaser and statistics blocks VERIFIED

LICENSING NOTE — ACT ON THIS IMMEDIATELY

The \$99 MATLAB and Simulink Student Suite **does not include** Deep Learning Toolbox, Computer Vision Toolbox, Medical Imaging Toolbox or SimEvents **VERIFIED**. Also note that **Parallel Computing Toolbox** is what enables GPU training and is not named in the PS tool list — omit it from your trial selection and you will train on CPU without understanding why everything is slow. See §14.1 for the licensing path.

5. Component 1 — Image Quality Assessment and Enhancement

Discharges R1.1, R1.2, R1.3. This is the component to build first: it requires no training, no GPU and no labelled data, and it is the PS's own stated differentiator against existing solutions.

5.1 Why this comes first

A deployed Google DR screening system in Thai clinics had a substantial fraction of images rejected for quality in field conditions, a finding reported in a human-centred evaluation study at CHI 2020.

UNVERIFIED — CONFIRM THE EXACT REJECTION PERCENTAGE BEFORE CITING The general lesson is robust and is exactly what the PS asserts: **models that perform well on curated datasets degrade in the field primarily because of image quality, not because of the model.**

Building the gate first also has a practical benefit for a four-day demo: it is entirely classical computer vision, so it produces a working, demonstrable stage on day one while the classifier is still training.

5.2 Quality metrics (R1.1)

All metrics are computed **inside the field-of-view mask only**. This matters: the black surround of a fundus image would otherwise dominate every intensity statistic and make the measures meaningless.

Step 1 — Field-of-view mask

Threshold the image to separate the illuminated circular retina from the black border, take the largest connected component, fill holes, and fit a circle. The mask is reused by every downstream stage, so compute it once and cache it.

```
% Approximate structure; tune thresholds per camera
gray = im2gray(I);
bw = imbinarize(gray, 0.05);           % retina vs black surround
bw = imfill(bw, 'holes');
bw = bwareafilt(bw, 1);               % largest component only
stats = regionprops(bw, 'Centroid', 'EquivDiameter', 'BoundingBox');
fovMask = bw;
fovRadius = stats.EquivDiameter / 2;
```

Step 2 — Quantitative quality features

Feature	Targets	Method
Focus	Blur, defocus	Variance of the Laplacian computed within the FOV mask. Also compute Tenengrad (sum of squared Sobel gradient magnitudes) as a second, less noise-sensitive estimator. Low values indicate blur.
Illumination uniformity	Uneven flash, vignetting	Estimate the background by heavy Gaussian or median filtering, then compute the coefficient of variation of that background across the four ETDRS quadrants. High variation indicates uneven illumination.
Global exposure	Under/over-exposure	Mean and percentile intensities within FOV; fraction of saturated pixels (blown highlights) and fraction of near-black pixels.
Contrast	Washed-out images	RMS contrast and Shannon entropy of the green channel within FOV. The green channel carries most retinal lesion contrast. ESTABLISHED

Feature	Targets	Method
Field definition	Wrong field, off-centre	Requires that the optic disc is detected (§6.2) and that the estimated fovea position lies within the FOV. A macula-centred field is the screening standard.
Artefacts	Dust, lens flare, eyelashes	Detect anomalously bright small blobs and high-frequency arc structures near the FOV boundary.

5.3 Quality classifier

The features above feed a small classifier producing three classes: **gradable**, **borderline (enhance and retry)**, **ungradable (reject)**.

Use `fitcecoc` or a decision tree from Statistics and Machine Learning Toolbox rather than a neural network. Three reasons, and they are all worth being able to say out loud:

1. The feature set is small and hand-designed, so a deep model has nothing to add.
2. A decision tree is itself interpretable, which is consistent with the project's thesis. The quality gate can state *which* feature caused a rejection, and that is precisely what generates the recapture message in §5.5.
3. It trains on a few hundred hand-labelled examples in seconds on CPU.

Labels. Hand-label roughly 300–400 APTOS images into the three tiers. This is a two-hour task for two people and needs no clinical training — "can you see the retina clearly enough to look for small red dots?" is the operative question. Record the labelling protocol in the repository so it is reproducible.

5.4 Enhancement (R1.2)

Applied only to borderline images. Order matters.

#	Operation	Detail
1	Illumination normalisation	Estimate the slowly varying background with a large-kernel filter and subtract it, then re-centre. Removes vignetting and uneven flash without touching lesion-scale detail.
2	CLAHE	<code>adapthisteq</code> applied to the green channel, or to the L channel in Lab space to preserve colour relationships. Contrast-limited so it does not amplify noise into false lesions.
3	Denoising	<code>imnlmfilt</code> (non-local means). Applied conservatively — over-denoising erases microaneurysms, which are exactly the small dark dots a denoiser wants to remove.

SELF-INFLICTED DOMAIN SHIFT — THE MOST LIKELY SILENT BUG IN THIS COMPONENT

If enhancement is applied at inference but not during training, you have created a distribution mismatch with your own hands, and the model will underperform for a reason no metric will name. **Rule: the enhancement decision is part of the pipeline, and the pipeline is identical at training and inference time.** Both must run the same gate and apply the same enhancement to the same category of image. Encode this by having exactly one preprocessing function, called from both paths. Never two.

5.5 Recapture feedback (R1.3)

Rejection alone is not what the PS asks for. It asks for rejection "with recapture feedback" — the system must tell the operator what to fix. Because the quality classifier is a tree over named features, the dominant failing feature is directly recoverable and maps to an instruction:

Dominant failing feature	Operator message	Corrective action
Low Laplacian variance	"Image out of focus"	Refocus; ask patient to fixate on the target
Low mean intensity	"Image too dark"	Increase flash intensity; check pupil dilation
High saturated fraction	"Image over-exposed"	Reduce flash intensity
High quadrant illumination CV	"Uneven lighting"	Re-align camera to the eye; check working distance
Optic disc not detected / fovea outside FOV	"Macula not centred"	Re-aim; the macula should be near the image centre
Bright arc near boundary	"Lens flare or eyelash"	Ask patient to open the eye wider; clean the lens

DEMO VALUE

This table is one of the strongest live-demo moments available to you, and it costs almost nothing to build. Feed the system a deliberately defocused image and watch it say "Image out of focus — refocus and retake" instead of confidently grading noise. It directly demonstrates the PS's own complaint that existing systems "fail with variable image quality," and it visibly does something about it.

6. Component 2 — Retinal Structure Segmentation

Discharges R2.1–R2.6. This is the component that makes lesion-level explanation possible, and it is also where the honest limits of the available data must be stated.

6.1 Two-track strategy

Every structure below is addressed on two tracks, and this is a deliberate risk-management decision rather than indecision:

Track	Nature	Purpose
Track A — Classical	Morphology, filtering, geometric priors. No training.	Works on day one. Provides a functioning demo before any model has converged, and remains the fallback if a segmentation model fails to train usefully on small data.
Track B — Learned	Patch-based semantic segmentation networks trained on IDRI and DRIVE.	Higher ceiling. The version presented at the grand finale.

Track A is not a placeholder to be embarrassed about. Classical morphological microaneurysm detection is a genuine, long-established line of work in retinal image analysis, and having a non-learned baseline to compare against is itself part of the ablation R6.2 demands.

6.2 Optic disc and fovea localisation (R2.1)

Track A. The optic disc is the brightest large structure in a normal fundus image and is the point of vessel convergence. Locate candidate bright regions of approximately disc size within the FOV, then disambiguate using vessel density, since vessels radiate from the disc.

The fovea is then found geometrically. It sits approximately 2.5 disc diameters temporal to the disc centre, roughly on the horizontal axis through the disc. **ESTABLISHED** Refine by searching that neighbourhood for the darkest avascular region.

Track B. IDRI provides optic disc masks and fovea centre coordinates, so a small U-Net for the disc and a coordinate-regression head for the fovea are both directly trainable.

Why we need these at all. Three downstream uses, and a judge may well ask: field-definition checking in the quality gate (§5.2); quadrant definition for the 4-2-1 rule, which is defined relative to the disc-fovea axis (§8.5); and exudate disambiguation, since the optic disc is bright and yellow and is the classic false positive for hard exudate detection.

6.3 Vessel segmentation (R2.2)

Dataset: DRIVE, 40 images with expert vessel annotations, conventionally split 20 train / 20 test.

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Approach: patch-based U-Net on the green channel. Forty images is very few for whole-image training, but extracting overlapping patches yields tens of thousands of training samples, which is ample. This is the standard approach for DRIVE and is why a 40-image dataset remains usable.

Metrics: sensitivity, specificity and AUC against the expert masks, which are the conventional DRIVE benchmark measures and let us compare to published results as R6.1 requires.

Downstream uses: venous beading assessment for the 4-2-1 rule; vessel masking to suppress false lesion detections, since dark vessel segments are readily mistaken for haemorrhages; and neovascularisation features (§6.6).

6.4 Microaneurysm detection (R2.3) — the hard one

SET EXPECTATIONS HONESTLY HERE

The PS asks for "sub-pixel microaneurysm detection." Take that phrase seriously rather than treating it as flourish: it is telling you that MAs are at or below the effective resolution of a naively downsampled pipeline. IDRiD's segmentation subset provides on the order of a few dozen images with MA masks. **UNVERIFIED — CONFIRM EXACT COUNTS ON DOWNLOAD** This is a very small training set for a very small object. **Plan for this to be the component that most nearly fails, and build Track A so that it cannot fail completely.**

Track A — morphological candidate detection

The classical pipeline, which needs no training data at all:

1. Take the inverted green channel, where MAs appear as small bright peaks.
2. Apply morphological top-hat with **linear structuring elements at multiple orientations**. Elongated structures (vessels) survive at least one orientation and are suppressed; small round structures (MAs) do not, and are retained. This is the key trick and it exploits the shape difference directly.
3. Threshold to obtain candidates.
4. Extract per-candidate features: area, eccentricity, peak intensity, local contrast, distance to nearest vessel.
5. Classify candidates as MA or not with a small model from Statistics and Machine Learning Toolbox.

Track B — patch-based segmentation at native resolution

Decision	Choice	Reason
Input scale	Native resolution patches (e.g. 512×512 crops from the full ~4288×2848 image)	Downsampling the whole image destroys the target. This is non-negotiable, see §3.2.
Loss function	Dice + focal, or Tversky weighted toward recall	MA pixels are a vanishingly small fraction of all pixels. Plain cross-entropy converges to predicting all-background and reports excellent loss while detecting nothing.
Primary metric	Area under the precision-recall curve	With extreme class imbalance, ROC-AUC looks impressive and means little. AUPR is the metric used for the IDRiD lesion segmentation benchmark. UNVERIFIED — CONFIRM
Augmentation	Flips, rotations, modest colour/brightness jitter	Compensates for tiny training set. Avoid heavy blur or noise augmentation, which destroys the target class.
Sampling	Oversample patches containing lesions	Random patches from a fundus image are almost all lesion-free.

6.5 Haemorrhages, hard exudates, soft exudates (R2.4, R2.5)

Same patch-based approach as §6.4 Track B, trained on the corresponding IDRiD masks. These are substantially easier than MAs because the objects are larger.

Class	Discriminating cue	Principal false-positive source and mitigation
Haemorrhage	Dark red, irregular boundary	Vessel segments. Mitigate by suppressing detections that overlap the vessel mask from §6.3.
Hard exudate	Bright yellow, sharp edges	The optic disc. Mitigate by excluding the disc region from §6.2.
Soft exudate	Pale, fluffy, indistinct edges	Hard exudates. Separate on edge sharpness and colour saturation.

Note the dependency structure this creates: vessel and disc segmentation are not independent deliverables, they are inputs that materially improve lesion precision. That interdependence is itself part of the R6.2 argument that the integrated pipeline beats any single technique — and it is measurable by running lesion detection with and without the vessel/disc suppression.

6.6 Neovascularisation (R2.6) — declared data gap

STATE THIS PLAINLY RATHER THAN PRETENDING

None of the four datasets named in the PS provides pixel-level neovascularisation annotations. IDRiD supplies image-level grades that include proliferative DR, and it supplies pixel masks for MA, HE, EX and SE — but not for NV. We therefore cannot train a supervised NV segmenter on the named data, and we will not claim to have done so.

What we do instead, and say we do:

1. **Image-level PDR detection via the classifier.** Level 4 is one of the five ICDR classes the grading model outputs, so proliferative disease is detected at image level even without lesion-level NV masks.
2. **Vessel-morphology proxy features.** From the vessel segmentation, compute abnormal branching density, tortuosity and calibre irregularity, particularly within one disc diameter of the optic disc where disc neovascularisation characteristically occurs. Report these as *supporting indicators*, explicitly not as a validated NV detector.
3. **Route to a human.** Any predicted Level 4 is escalated for human grading regardless of confidence. Proliferative DR is sight-threatening and comparatively rare, so the cost of always escalating it is small and the cost of an autonomous miss is severe.

This is the correct engineering answer *and* the correct presentation answer. A judge who knows these datasets knows there are no NV masks in them. A team that claims a trained NV segmenter is either mistaken or has quietly used data they have not declared. A team that names the gap, explains the mitigation, and escalates the risky class is demonstrating exactly the "clinical validation rigor" the PS says the field lacks.

7. Component 3 — DR Severity Grading

Discharges R3.1, R3.2, R3.3, R4.3. The classifier is the least novel part of this project and the part most teams spend all their time on. Budget accordingly.

7.1 Backbone selection

Decision	Choice	Reason
Architecture	ResNet-50 initially; EfficientNet as a comparison if time allows	Pretrained and available through <code>imagePretrainedNetwork</code> . Well understood, and <code>gradCAM</code> works cleanly on its final convolutional block.
Initialisation	ImageNet pretrained weights, fine-tuned	~6,000 total images across the named datasets is far too few to train from scratch.
Input resolution	448×448 minimum, 512×512 preferred	The default 224×224 destroys microaneurysm evidence (§3.2). This is the single most consequential hyperparameter in the project.
Batch size	8–16, tuned to fit 6 GB VRAM	Constrained by the RTX 3050. If unstable, reduce batch size before reducing resolution — resolution is worth more here.

7.2 Preprocessing

1. **Crop to the FOV bounding box** using the mask from §5.2, removing the black surround. Without this, a large fraction of every input tensor is uninformative black pixels.
2. **Resize to a fixed square** at the chosen resolution.
3. **Local colour normalisation.** Subtracting a heavily blurred version of the image from itself, then re-centring, substantially normalises illumination and colour differences between cameras. This technique is strongly associated with top solutions in the original Kaggle DR competition.

UNVERIFIED ATTRIBUTION — DESCRIBE THE METHOD, DO NOT ATTRIBUTE IT TO A NAMED INDIVIDUAL WITHOUT CHECKING

4. **Per-channel standardisation** using training-set statistics.

Steps 1–4 are implemented in a *single shared function* used by both the training and inference paths, for the reason given in §5.4.

7.3 Output head (R3.1)

DR grades are **ordinal**, not categorical: confusing Level 0 with Level 4 is a far worse error than confusing Level 2 with Level 3. A plain 5-way softmax treats all confusions as equal and throws that structure away.

Option	Assessment
5-class softmax	Simplest. Ignores ordinality. Adequate baseline.
Regression + learned thresholds	Respects ordinality; historically strong on this task. Slightly awkward to interpret probabilistically, which matters for R4.3 calibration.
Ordinal (cumulative) encoding — four binary heads predicting $P(\text{grade} > 0)$, $P(\text{grade} > 1)$, $P(\text{grade} > 2)$, $P(\text{grade} > 3)$	Recommended. Respects ordinality, and the head $P(\text{grade} > 1)$ is the referable-DR probability — the exact quantity the PS sets its targets on. The primary endpoint becomes a direct model output rather than a post-hoc sum.

Ship the softmax baseline first, then the ordinal head, and report both. The comparison is a legitimate ablation contributing to R6.2.

7.4 Class imbalance

APTOS is markedly imbalanced, with Level 0 the dominant class and Levels 3 and 4 sparse.

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Handle it with, in order of preference: class-weighted loss; oversampling of minority classes in the training datastore; and augmentation applied more aggressively to minority classes. Do *not* attempt to fix imbalance by discarding Level 0 images — the deployment distribution is imbalanced too, and the model needs to see it.

MANDATORY LOGGING

Every training run in this project prints **per-class recall and the full confusion matrix** at every validation epoch. This is a hard requirement, not a nicety. A model that collapses to predicting the majority class produces a healthy-looking loss curve and a respectable accuracy figure, and raises no error at all. Per-class recall is the only cheap signal that catches it. See §13.4.

7.5 Uncertainty estimation

Required for the deferral logic in §4.2. Two viable approaches:

Method	Cost	Notes
MC dropout	N forward passes at inference ($N \approx 20$)	Keep dropout active at inference; the spread across passes estimates uncertainty. Single model, so no extra training cost. Recommended for the 6 GB constraint.
Deep ensemble	3–5× training and storage	Generally better-calibrated uncertainty, but multiplies both training time and the on-device model footprint, which conflicts with offline PHC deployment.

Uncertainty score: predictive entropy of the mean predictive distribution across passes. Cases above a threshold are escalated. The threshold is not guessed — it is chosen from the risk-coverage curve (§11.4) and then fed into the SimEvents model (§9.5) as the parameter that determines grader workload.

7.6 Probability calibration (R4.3)

The PS asks for "calibrated confidence scores." Raw softmax output is *not* calibrated — modern neural networks are systematically overconfident. Presenting a softmax value as a confidence is one of the most common unforced errors in student ML projects, and it is easy to get right.

METHOD

1. Hold out a calibration split, distinct from both the training set and the test set.
2. Fit a single temperature parameter T that minimises negative log-likelihood on that split, dividing the logits by T before the softmax.
3. Report **Expected Calibration Error** before and after, and include a **reliability diagram** in the results.

Temperature scaling is a one-parameter fit, takes minutes, and produces a defensible answer to "what does 0.87 mean?" — namely that among cases where the system reports 0.87, close to 87% genuinely are referable. That sentence, backed by a reliability diagram, is a strong judging moment.

8. Component 4 — Explainability Module

Discharges R4.1, R4.2, R4.4, and supports R6.2 and R6.3. This is the heart of the problem statement and the section Rithviksai should know best.

8.1 Three channels, deliberately

Read R4.1 and R4.2 together: the PS asks for Grad-CAM attention maps *and* lesion-level evidence correlated with clinical criteria. It asks for both because they are different kinds of explanation with different failure modes.

Channel	Question answered	Faithful to	Weakness
1. Grad-CAM (post-hoc)	"Where in the image did the network's decision come from?"	The model	Coarse spatial resolution; cannot localise microaneurysms. Explains the model whether or not the model is right.
2. Lesion evidence (inherently interpretable)	"What clinical findings are present, where, and how many?"	The clinical criteria	Depends on lesion detector quality; can miss findings the classifier picked up on.
3. ICDR rule trace	"Which published criterion produces this grade from those findings?"	The guideline	Rule-based grade may disagree with the CNN grade.

Their disagreements are not a defect of the design. They are the signal (§8.6).

8.2 Channel 1 — Grad-CAM and post-hoc saliency (R4.1)

MATLAB's Deep Learning Toolbox provides exactly three image-explanation functions: `gradCAM`, `occlusionSensitivity` and `imageLIME`. VERIFIED Anything beyond these must be implemented manually with `dlfeval` and `dlgradient`.

Function	Cost	Use in this project
<code>gradCAM</code>	One backward pass	Primary, mandated by R4.1. Fast enough for the live UI.
<code>occlusionSensitivity</code>	Many forward passes	Secondary. With a small mask size it gives finer localisation than Grad-CAM but is far slower. Use for offline analysis and the comparison study, not the live path. Supports non-classification tasks from R2024a. VERIFIED
<code>imageLIME</code>	Many forward passes	Secondary. Superpixel-based; produces segment-level importance. Included in the method comparison.

8.3 The Grad-CAM resolution limitation — know this number

Grad-CAM computes its map at the spatial resolution of the chosen convolutional layer, then upsamples. For ResNet-50, the final convolutional block has a stride of 32 relative to the input. **At 448×448 input, the Grad-CAM map is 14×14.** Each cell of that map therefore covers roughly a 32×32-pixel region of the resized input.

A microaneurysm at that scale occupies a fraction of one cell. **Grad-CAM at the final layer physically cannot localise a microaneurysm.** This is not a tuning problem; it is arithmetic.

What we do about it, and what we say about it:

1. **Compute Grad-CAM at multiple depths.** Earlier layers give finer maps (lower stride) but weaker class specificity. Present the trade-off explicitly rather than silently picking one. Selecting the layer is a `gradCAM` argument, so this is inexpensive.
2. **Do not oversell it.** Grad-CAM answers "which region drove this decision," and for large lesions such as exudates and haemorrhages it does that well. Report it as regional attention, not lesion localisation.
3. **Supply channel 2 for the fine detail.** This is precisely why the PS asks for lesion-level evidence in the same breath.
4. **Measure the gap rather than asserting it** (§11.7).

Being able to state this limitation crisply is worth more in front of a MathWorks judge than any accuracy figure in the deck. It demonstrates that you understand the tool rather than having invoked it.

8.4 Channel 2 — lesion-level evidence (R4.2)

From §6, we have per-lesion detections. Convert them into clinical evidence:

- Divide the retina into the four **quadrants** defined relative to the optic disc–fovea axis (superior-temporal, inferior-temporal, superior-nasal, inferior-nasal). This is the spatial partition the ICDR criteria are written against, so it is not an arbitrary choice.
- Count detections per lesion class per quadrant.
- Record the distance of each hard exudate from the fovea, since exudates near the macula carry different clinical significance.
- Render as coloured overlays with a legend and a counts table.

8.5 Channel 3 — the ICDR rule trace

Apply the §3.3 criteria to the counts from §8.4 and emit the reasoning as text. This is the "correlated with clinical criteria" half of R4.2, made literal.

LESION EVIDENCE

Microaneurysms 17 (ST 6 · IT 5 · SN 3 · IN 3)
Haemorrhages 4 (ST 2 · IT 2 · SN 0 · IN 0)
Hard exudates 2 (nearest to fovea: 1.4 disc diameters)
Soft exudates 0
Venous beading not detected
Neovascularisation not assessed - see limitations

ICDR RULE TRACE

Level 1 requires microaneurysms only exceeded
Level 3 (4-2-1) requires >20 haemorrhages in 4 quadrants
/ beading in 2+ quadrants / IRMA not met
=> More than MA only, less than severe NPDR
=> LEVEL 2 - MODERATE NPDR -> REFERABLE

CNN GRADE Level 2 (calibrated confidence 0.88)

AGREEMENT Rule path and CNN agree

WHY THIS BEATS A HEATMAP IN FRONT OF A CLINICIAN

A heatmap says "the model looked here." This says "there are 17 microaneurysms and 4 haemorrhages distributed thus, which under the published ICDR criterion is Moderate NPDR, which is referable." The second is checkable. An ophthalmologist can look at the image and verify the counts. That is what makes validation in under 30 seconds (R4.5) realistic, and it is what we measure in §12.

8.6 The agreement check and the unexplained-prediction flag

With three channels we can cross-examine the system's own output. Four states:

State	Condition	Action
Concordant	CNN grade and rule-path grade agree; Grad-CAM peaks coincide with detected lesions	Proceed with the automated decision. Report high explanation confidence.
Under-detected	CNN says referable; lesion detector finds little or nothing	Escalate. Either the detector missed findings or the classifier is keying on something that is not disease.
Over-detected	Lesion detector finds substantial pathology; CNN says not referable	Escalate. Favours safety, since the lesion path is the clinically grounded one.
Spatially inconsistent	Grades agree, but Grad-CAM attention falls outside all detected lesions	Flag in the report as reduced explanation confidence. Consider escalation.

The under-detected state is the important one. Retinal classifiers are known to be able to exploit non-pathological image properties, and a model that has learned a camera artefact will still produce a confident prediction and a plausible-looking heatmap. Standard post-hoc XAI presents that heatmap as a justification, which is worse than offering no explanation at all. Our pipeline detects the condition and declines to decide.

8.7 Automated annotated report (R4.4)

One page, generated per patient, designed to be read by a health worker and checked by a remote grader.

Element	Content
Header	Patient ID, date, eye (L/R), camera, PHC identifier

Element	Content
Quality verdict	Tier, and whether enhancement was applied
Images	Original; enhanced; lesion overlay; Grad-CAM overlay — four panels
Evidence block	The lesion counts table and ICDR rule trace from §8.5
Decision	Auto-clear / Refer / Escalate, with calibrated confidence
Agreement status	Concordant, or the specific disagreement detected
Footer	"Screening aid. Not a diagnosis. Requires clinician confirmation." — non-negotiable, see §17.4

Implementation. MATLAB Report Generator is a separate product and is *not* among the six the PS names, so do not build a dependency on it. Compose the report as a figure or tiled layout and write it out with `exportgraphics` to PDF, appending pages as needed. This keeps the deliverable inside the declared toolbox set.

9. Component 5 — SimEvents District Capacity Model

Discharges R5.1, R5.2. No GPU, no dataset, no machine learning. Buildable in parallel with everything else, from day one. This is the highest return-per-hour component in the entire problem statement.

9.1 Why this matters more than its size suggests

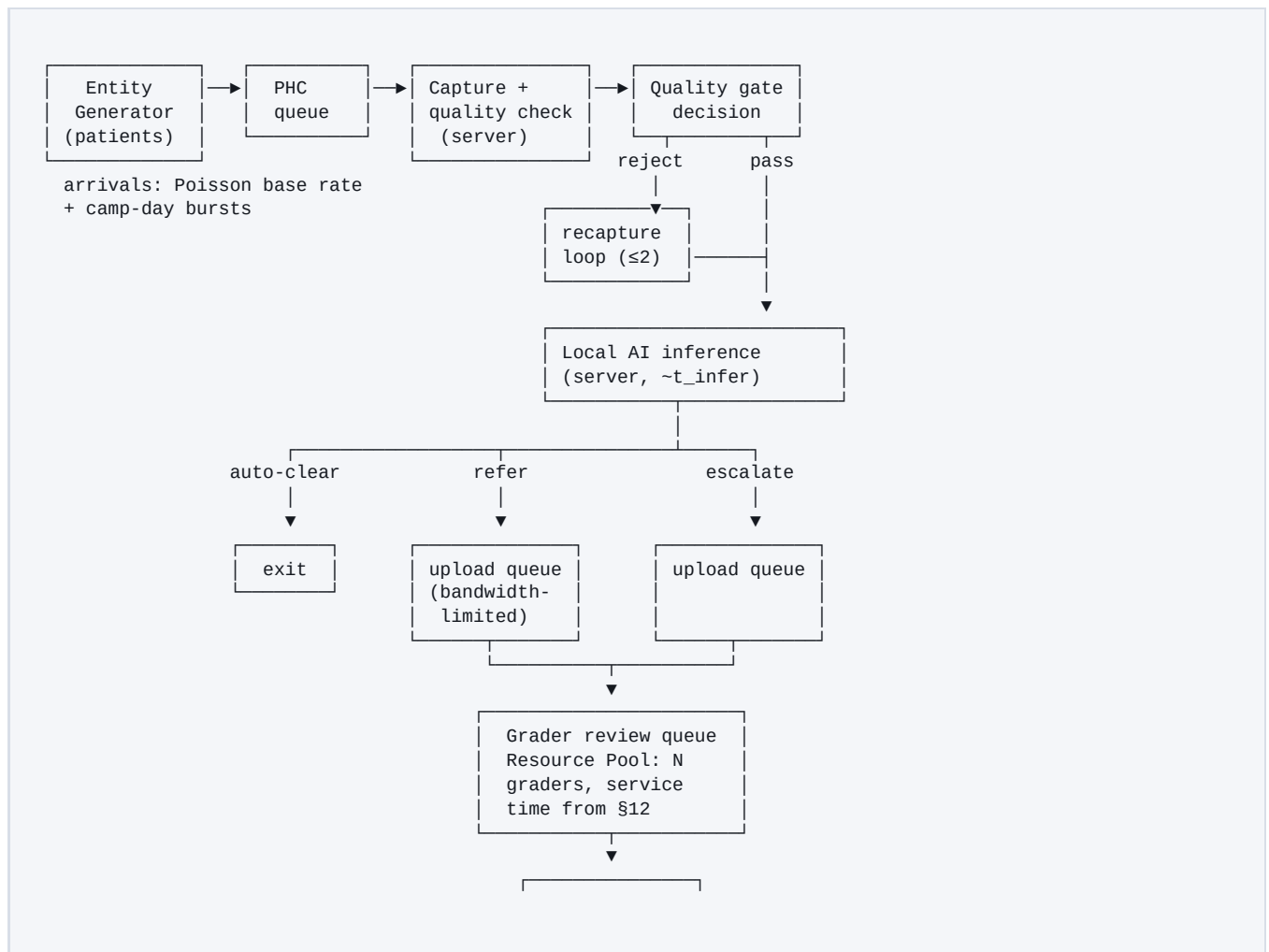
Requirement 5 is a fifth of the problem statement. It is also the part that most teams will skip, reduce to a static bar chart, or discover on the morning of the presentation. It is the MathWorks-specific portion of a MathWorks problem statement, and it is the part a MathWorks judge is personally best placed to evaluate.

It also closes a loop that nothing else in the pipeline closes. Everywhere else, the classifier's operating point is a number we choose to satisfy a target. Here it becomes an *input to a capacity model* with real consequences: raising sensitivity produces more false positives, which consume more grader hours, which lengthen turnaround for everyone. That trade-off is quantifiable, and quantifying it is precisely the "optimise resource allocation" that R5.2 asks for.

9.2 Tooling

SimEvents supplies the required block types directly: Entity Generator, Queue, Server, Entity Input/Output Switch for routing, and Resource Pool / Resource Acquirer / Resource Releaser for modelling a limited pool of human graders, together with statistics collection on throughput, latency and utilisation. **VERIFIED**

9.3 Model structure (R5.1)



9.4 Parameters

Every parameter is a named variable in a configuration file, never a number typed into a block dialog. This is what makes §9.5 possible.

Parameter	Indicative value	Source / note
Annual screening volume	100,000+	Set by R5.2
Arrival process	Poisson base + camp-day bursts	NCD screening camps produce highly bursty arrivals; a flat rate would understate peak queueing badly
Number of PHCs	Sweep parameter	Determines parallelism at the capture stage
Capture time per patient	Sweep, order of minutes	Includes both eyes and patient handling
Quality rejection rate	Sweep; measured from §5 on real data	Links C1 to C5. Our own measured rejection rate becomes this model's input.
Recapture attempts allowed	2	After which the patient is referred for manual examination
AI inference time	Measured on target hardware	Measure it; do not assume it
Upload bandwidth	Sweep	Service time modelled as file size / available bandwidth. Rural connectivity is the binding constraint here.
Connectivity availability	Sweep (e.g. duty cycle)	Drives the store-and-forward queue depth
Number of graders	Primary sweep variable	The scarce resource. The PS states ~1 ophthalmologist per 100,000 rural population.
Grader service time	From the reader study, §12	Links C4 to C5. The measured ~30 s target from R4.5 is this model's service-time distribution.
AI sensitivity / specificity	From §11.2	Links C3 to C5. Determines referral and false-positive volume.
Deferral threshold	From §7.5	Determines escalation volume

NOTE THE FOUR ARROWS BACK INTO THE REST OF THE PROJECT

Quality rejection rate, grader service time, model sensitivity/specificity, and deferral threshold are all *measured outputs of components 1, 3 and 4* feeding into component 5 as inputs. This is what makes the Simulink model an integrated part of the system rather than a decorative side-exhibit, and it is a strong thing to be able to point at during judging.

9.5 Optimisation experiments (R5.2)

Outputs collected: end-to-end turnaround distribution, grader utilisation, queue lengths over time, throughput, and the fraction of patients whose result is issued within the target window.

#	Experiment	Question answered
E1	Sweep grader count at fixed volume	What is the minimum number of graders needed to keep turnaround within target for 100,000 patients per year? The headline resource-allocation answer.
E2	Sweep AI operating point	The interesting one. Raising sensitivity from 90% to 95% increases false positives, which increases grader load. Plot grader-hours per year against sensitivity to expose the trade-off.
E3	Sweep deferral threshold	How much human capacity does the escalation path consume, and what accuracy does it buy? Directly quantifies the cost of the safety mechanism in §8.6.

#	Experiment	Question answered
E4	Sweep bandwidth and connectivity duty cycle	At what connectivity does upload, rather than grader capacity, become the bottleneck? Determines whether local inference is necessary or merely convenient.
E5	Toggle the quality gate	What happens to grader load if ungradable images are forwarded instead of rejected? Feeds R6.2 directly: demonstrates the quality gate's system-level value in grader-hours, not just in accuracy.
E6	Bursty vs smooth arrivals	How much does camp-day bunching degrade turnaround, and does buffering fix it?

Headline deliverable from this component: a Pareto curve of screening sensitivity against annual grader-hours, with the PS's >90% sensitivity target marked on it, and a stated minimum viable grader count for a district of 100,000 screenings per year. That is a genuine operations-research result derived from your own measured model parameters, and no other team will have one.

10. Data Strategy and Validation Protocol

This section is the project's answer to the PS's charge that existing solutions "lack clinical validation rigor." It is also the section most likely to be undermined by an innocent shortcut, so it is written as rules rather than suggestions.

10.1 Datasets

Dataset	Size	Labels	Origin	Role here
APTOS 2019	~3,662 train	ICDR 0–4, image level	Aravind Eye Hospital, India	Primary training set. Indian population and Indian cameras — the closest available proxy for the deployment setting.
IDRiD	~516 graded; smaller subset with masks	ICDR grades; pixel masks for MA, HE, EX, SE ; optic disc masks; fovea coordinates	Nanded, India	Lesion supervision and XAI ground truth. The only named dataset with pixel-level lesion masks, which makes it the basis for §11.7.
DRIVE	40	Vessel masks	Netherlands	Vessel segmentation training and benchmarking
Messidor-2	~1,748 images	DR grades	France	SEALED EXTERNAL TEST SET. Different country, population and cameras. See §10.4.

UNVERIFIED — all counts above are believed correct but must be confirmed against the actual downloads and recorded here by the data owner. Do not quote them in the submission until checked.

ACCESS LEAD TIMES — START TODAY

Messidor-2 is distributed under a licence agreement rather than by direct download, and IDRiD is hosted on IEEE DataPort. Both can take days to clear. **Request access on day one.** A sealed external test set you cannot obtain until December is not a validation strategy. If Messidor-2 access does not arrive in time, the fallback is a held-out, patient-disjoint IDRiD split as the external set, clearly labelled as a weaker substitute because it shares an Indian origin with the training data.

10.2 Split discipline

Rule	Reason
Patient-level splits, never image-level	Left and right eyes of one patient are correlated. If one eye is in training and the other in test, the test score is inflated and the inflation is invisible. This is the single most common silent leak in retinal ML.
Four splits, not three	Train / validation / calibration / test. Temperature scaling (§7.6) must be fitted on data used for nothing else, or the reported calibration is optimistic.
Fixed seed, split file committed	The split is generated once, written to a CSV, and that CSV is committed. Never regenerate the split at run time.
Stratified by grade	Levels 3 and 4 are sparse; unstratified random splitting can leave a validation fold with almost none.

10.3 Data provenance log

Maintain `data/PROVENANCE.md` recording, for each dataset: source URL, download date, licence terms, file count and total size, SHA-256 of the archive, and any preprocessing applied before use. This takes fifteen minutes and answers, instantly and credibly, any judge question about where the data came from and whether you were entitled to use it.

10.4 The sealed test set — the core rigour mechanism

PROTOCOL

1. On download, Messidor-2 is placed in an archive that is **encrypted or otherwise access-controlled**, and one named team member holds the key. It is not extracted into the working tree.
2. All development — architecture selection, hyperparameter tuning, threshold selection, calibration — happens using train / validation / calibration splits only.
3. When the team judges the system complete, the operating point is **frozen and recorded in this document** with a date, along with the validation-set metrics that justified it.
4. Only then is the sealed set opened, and the frozen pipeline run over it **exactly once**.
5. Whatever comes out is reported. **If external performance falls below the PS targets, that result is reported and analysed, not tuned away.**
6. If any change is made to the pipeline after opening the seal, the resulting number ceases to be an external validation result and must be reported as such.

Two reasons this is worth the discipline. The scientific one: any threshold chosen with knowledge of the test set makes the reported figure an optimistic estimate of deployed performance, and screening decisions affect eyesight. The competitive one: the PS explicitly criticises existing solutions for lacking validation rigour. A team that can describe this protocol, and show the frozen operating point with a date preceding the test run, has answered the problem statement's central complaint with a procedure rather than a claim.

10.5 Expected domain shift, stated in advance

We predict, before running the experiment, that performance on Messidor-2 will be **lower** than on the internal APTOS test split. Recording the prediction here, in advance, is part of the protocol.

Shift factor	Nature
Population	Indian (APTOS) vs French (Messidor-2) — different fundus pigmentation distributions
Camera and optics	Different manufacturers, fields of view, resolutions and colour rendering
Acquisition protocol	Different dilation practice, field definition and operator training
Label process	Different grading protocols and grader pools; APTOS grades are single-grader, so label noise is present in training

A drop is the expected, honest outcome and is well documented across the medical imaging literature. Reporting it, decomposing it, and proposing mitigation (colour and illumination normalisation, quality-tier stratification, and camera-specific recalibration on a small local sample) is a stronger result than an implausibly flat external number.

11. Evaluation Framework

Discharges R3.2, R3.3, R6.1, R6.2. Build this before you build the models. An evaluation harness written after the fact gets written to flatter whatever the model already does.

11.1 Reporting policy

Never report	Always report instead
Bare accuracy	Sensitivity and specificity at the frozen operating point, with 95% confidence intervals
A single number with no interval	Point estimate with a Wilson score interval; state n
Best epoch selected on the test set	Epoch selected on validation; test touched once
Metrics on a filtered "clean" subset only	Metrics on the full set, plus a stratified breakdown by quality tier
Softmax output described as "confidence"	Temperature-scaled probability, with ECE and a reliability diagram

11.2 Primary endpoint (R3.2, R3.3)

Referable DR = predicted ICDR Level ≥ 2 . With the ordinal head of §7.3, this is the direct output $P(\text{grade} > 1)$.

THRESHOLD SELECTION PROTOCOL

1. On the **validation** split, sweep the decision threshold across its range.
2. Select the threshold that achieves sensitivity ≥ 0.90 , and among those, maximises specificity.
3. **Freeze it. Record the value and the date in the table below.**
4. Apply the frozen threshold unchanged to the internal test split and to the sealed external set.

Frozen threshold value	<i>to be recorded on freeze</i>
Freeze date	<i>to be recorded on freeze</i>
Validation sensitivity / specificity at freeze	<i>to be recorded on freeze</i>
Frozen by	<i>name</i>

11.3 Full metric set

Metric	Applies to	Why
Sensitivity, specificity at frozen threshold, with 95% CI	Referable DR	Primary endpoint. Directly tests R3.2 and R3.3.
PPV and NPV at screening prevalence	Referable DR	What the number actually means to a patient. Prevalence-dependent, so state the prevalence assumed.
ROC curve and AUC	Referable DR	Threshold-independent summary; comparable to published benchmarks (R6.1)
Precision–recall curve and AUPR	Referable DR; all lesion segmentation	More informative than ROC under class imbalance

Metric	Applies to	Why
Quadratic weighted kappa	5-class grade	The standard ordinal agreement metric for DR grading and the APTOS competition metric. Penalises distant confusions more than adjacent ones. UNVERIFIED — CONFIRM AS THE APTOS METRIC
Full confusion matrix and per-class recall	5-class grade	Exposes majority-class collapse. Mandatory in every run.
Expected Calibration Error; reliability diagram; Brier score	Probabilities	Discharges R4.3
Risk–coverage curve	Deferral policy	Accuracy as a function of the fraction of cases the system handles autonomously. Sets the deferral threshold and feeds §9.5 E3.
Stratification by quality tier	All of the above	Directly addresses the PS's "fail with variable image quality" charge
Stratification by dataset / camera	All of the above	Exposes domain shift (§10.5)
Inference latency and model size	Deployment	Feeds §9.4; determines offline feasibility

11.4 Reporting internal versus external

Every headline result appears twice, side by side, in this shape:

Metric (referable DR)	Internal test APTOS held-out	External test Messidor-2, sealed	PS target
Sensitivity (95% CI)	result	result	> 90%
Specificity (95% CI)	result	result	> 85%
AUC	result	result	—
Quadratic weighted kappa	result	result	—
ECE	result	result	—

IF THE EXTERNAL NUMBER MISSES THE PS TARGET

Report it, then explain it using §10.5, then state what would close the gap: local recalibration on a modest sample of images from the deployment camera, quality-tier-specific thresholds, and additional training data from the target population. **This is a stronger submission than an unexplained near-perfect number**, because it is the only version a judge can actually believe, and because the PS asked for validation rigour rather than for a high score.

11.5 Comparison to published benchmarks (R6.1)

Compare against published results on the same datasets, being scrupulous about comparability. Note explicitly where a comparison is not like-for-like — different splits, different label sets, different endpoints. A carefully qualified comparison is credible; an unqualified leaderboard-style table is not.

Comparison	Against	Caveat to state
Referable DR sens/spec	Regulator-cleared systems (§3.5)	Different test populations and reference standards. Indicative context only, not a head-to-head.
5-class kappa	Published APTOS results	Public leaderboard splits differ from ours; ours is patient-level and stratified.

Comparison	Against	Caveat to state
Lesion segmentation AUPR	Published IDRiD segmentation results	Confirm whether the published split matches ours.
Vessel segmentation	Published DRIVE results	Standard 20/20 split, so this one is closest to like-for-like.
XAI localisation precision	Sparse BagNet 0.960 vs Guided Backprop 0.656 VERIFIED	Different dataset and lesion-matching definition. Reproduce the <i>comparison structure</i> on our data rather than claiming their numbers.

11.6 Ablation study (R6.2) — the requirement teams miss

R6.2 asks you to show the integrated pipeline beats any single-technique approach. That is an instruction to ablate. Run all configurations on the same held-out data with the same frozen threshold:

Cfg	Configuration	What its result establishes
A1	CNN only, no quality gate, no lesion evidence, no deferral	The single-technique baseline. This is what a typical competing solution is.
A2	CNN + quality gate	Isolates the contribution of C1
A3	Lesion detection + ICDR rules only, no CNN	The fully interpretable baseline. Expected to be weakest on raw accuracy — its value is as the interpretable reference and as the source of the agreement check.
A4	CNN + quality gate + deferral	Isolates the contribution of uncertainty-based escalation
A5	Full pipeline: quality gate + CNN + lesion evidence + agreement check + deferral	The integrated system. Should dominate on the safety-relevant metrics.

Report all five in one table. Note that A5's advantage will show most clearly not in raw accuracy but in **accuracy on the autonomously-handled subset** — that is the whole point of deferral, and it should be stated that way rather than apologised for. Comparing A5 against A1 at equal coverage is the fair comparison, and the risk–coverage curve is how you present it.

ARCHITECTURAL CONSEQUENCE

Configurations A1–A5 must be selectable from a config file, not by editing code. If ablation requires commenting out lines, it will be done once, badly, the night before. Build the pipeline with each stage independently switchable from day one (§13.3).

11.7 Explanation quality metrics — the differentiator

Almost every team will show heatmaps. Very few will **measure** them. IDRiD's pixel-level lesion masks make quantitative evaluation possible, and this is the main reason IDRiD is in our pipeline at all.

Metric	Definition and purpose
Lesion hit rate (pointing game)	Take the top- k % most salient pixels of the explanation map. What fraction falls inside a ground-truth lesion mask? Computed identically for Grad-CAM, occlusion sensitivity, LIME and the lesion-evidence channel. This is the head-to-head comparison the PS's R6.2 asks for, made numeric.
Deletion / insertion AUC	Progressively remove the pixels the explanation says matter most and track the predicted probability. A faithful explanation causes a fast drop (deletion) and a fast rise (insertion). Measures faithfulness to the model, which is distinct from agreement with clinical ground truth — report both, because an explanation can be faithful to a model that is wrong.

Metric	Definition and purpose
Sanity checks (parameter randomisation)	Progressively randomise the trained model's weights from the top layer down and recompute the explanation. If the map barely changes, the method is not explaining <i>this model</i> — it is largely an edge detector. A number of widely used saliency methods fail this test in the published literature. UNVERIFIED — CONFIRM THE REFERENCE BEFORE CITING WHICH METHODS FAIL Running this on your own maps is a strong rigour signal and costs one afternoon.
Channel disagreement rate	Fraction of cases in each of the four states of §8.6. Quantifies how often the unexplained-prediction flag fires, which is both a safety statistic and a direct input to §9.5 E3.
Clinician usefulness rating	From §12. Discharges R6.3 — "Grad-CAM outputs rated as clinically useful" — which cannot be discharged by any automated metric.

12. Clinical Reader Study Protocol

Discharges R4.5 and R6.3, and supplies the grader service-time distribution to §9.4. This is the highest-value-per-hour activity in the project and it requires one email.

12.1 Why it is worth the effort

R6.3 requires Grad-CAM outputs "rated as clinically useful," and R4.5 requires "ophthalmologist validation in under 30 seconds." **Neither can be discharged by an automated metric.** Both require a clinician looking at your outputs and a stopwatch.

The PLOS Digital Health study cited in §1.3 is essentially this experiment at scale, with six ophthalmologists. A twenty-case pilot with one ophthalmologist is entirely achievable by two students in an afternoon, and it converts two requirements from assertions into measurements.

12.2 Recruitment

You need one ophthalmologist for approximately two hours. A postgraduate resident in ophthalmology at any government medical college is an appropriate and reachable participant. Approach the department by email, state that this is a student project for a national hackathon, that no patient data is involved (all images are from public research datasets), and that you need two hours. Offer acknowledgement in the submission.

12.3 Design

Cases	20–30, stratified across ICDR levels, and deliberately including cases where the channels of §8.6 disagree
Reference standard	Dataset ground-truth grades
Arms	(1) image only; (2) image + AI grade; (3) image + AI grade + full explanation
Ordering	Counterbalanced across cases so that arm order cannot explain any effect; the same case is not seen twice in the same session
Primary measures	Time to decision (target: under 30 s, per R4.5); agreement with reference standard
Secondary measures	Explanation usefulness on a Likert scale; free-text comment on each; explicit flag for any explanation judged misleading

RECORD THE MISLEADING CASES ESPECIALLY CAREFULLY

An explanation rated *misleading* is a more valuable finding than one rated useful, and it is exactly the failure mode the unexplained-prediction flag (§8.6) exists to catch. If the clinician flags a case as misleading, check whether your agreement check had already flagged it. If it had, that is a validated safety mechanism and it belongs in the presentation. If it had not, that is a real finding about your system's limits, and reporting it is what rigour looks like.

12.4 Honest reporting

With a single reader and twenty cases this is a **pilot**, not a clinical validation study. Report it as a pilot: state n , state that it is one reader, give confidence intervals, and do not generalise. A correctly-labelled pilot is credible and impressive for a student team. A twenty-case study described as "clinical validation" invites a judge to dismantle it.

12.5 Fallback if no clinician is available

If nobody responds before the finale, do **not** fabricate ratings. Instead: present the protocol above as a designed, ready-to-run study; report the automated proxies from §11.7 in its place; and state plainly that clinical rating is pending. A well-specified unexecuted protocol is a legitimate deliverable. Invented clinician ratings are misconduct, and they are also easy to catch, because a judge will ask which hospital and which clinician.

13. Software Engineering

13.1 Repository layout

```
sih26038/
  config/
    default.json          % every hyperparameter and pipeline switch
    ablation_A1..A5.json  % §11.6 configurations
  data/
    PROVENANCE.md         % §10.3
    splits/               % committed CSV split files - never regenerated
    sealed/               % Messidor-2, access-controlled, §10.4
  src/
    +quality/             % C1
    +segment/             % C2
    +grade/               % C3
    +explain/             % C4
    +report/              % report generation
    +common/              % preprocessing shared by train and inference
  simulink/
    district_model.slx    % C5
    sweep_experiments.m   % §9.5 E1-E6
  eval/
    harness.m            % BUILD THIS FIRST
    metrics/
  app/
    ScreeningApp.mlapp     % App Designer UI
  tests/                 % matlab.unittest
  results/               % dated, never overwritten
  docs/
```

13.2 Conventions

Convention	Reason
MATLAB packages (<code>+folder</code>) for namespacing	Avoids function-name collisions as the codebase grows
<code>rng(seed)</code> at the top of every entry point	Reproducibility. A result you cannot reproduce is not a result.
Results written to a dated directory, never overwritten	Lets you compare runs and recover from a regression
Every run writes its full config alongside its results	Answers "what produced this number?" months later
Experiment Manager for hyperparameter sweeps	Reproducible, auditable, and a visible use of the MathWorks toolchain
<code>matlab.unittest</code> for the preprocessing and metric functions	Metric bugs silently invalidate every result downstream of them

13.3 Configuration-driven pipeline

Every stage must be switchable from configuration, because §11.6 requires running five pipeline variants and §9.5 requires sweeping parameters that originate in those stages.

```
{
  "pipeline": {
    "quality_gate": true,
    "enhancement": true,
    "lesion_evidence": true,
    "agreement_check": true,
    "deferral": true
  },
  "grading": {
    "backbone": "resnet50",
    "input_size": 448,
    "head": "ordinal",
    "batch_size": 12,
    "seed": 42
  },
  "operating_point": { "referable_threshold": null, "frozen_on": null }
}
```

13.4 Working with AI coding assistants

The build is driven through Claude Code by two people. That is a legitimate strategy and MATLAB supports it well, since `matlab -batch "statement"` runs headlessly, logs to stdout and stderr, and exits non-zero on failure VERIFIED — which is exactly the loop an agent needs. Two failure modes need explicit countermeasures.

Failure mode	Countermeasure
Silent training failure. A model that collapses to the majority class raises no error. The agent reports success. Loss curves look fine.	Build <code>eval/harness.m</code> before the first model. Make per-class recall and the confusion matrix print on every validation epoch. Add a unit test asserting that no class has zero recall on a smoke-test run. This is the single most important engineering practice in this document.
Inability to defend the work. At judging — and during the finale's live build with mentor checkpoints — you will be asked why a choice was made, not who typed it.	This document is the countermeasure. Every decision here carries its reason. Rithviksai reads §1–§3, §11 and §17 closely, and the "design rationale" notes in §5–§9.

LIKELY JUDGE QUESTIONS, AND WHERE THE ANSWER LIVES

Question	Answer
"Why 448×448 and not 224?"	§3.2 — microaneurysms are destroyed by downsampling
"Which layer did you use for Grad-CAM, and why?"	§8.3 — final conv block, 14×14 at 448 input; multi-depth comparison included
"How do you know your explanations are correct?"	§11.7 — lesion hit rate against IDRiD masks, deletion/insertion, sanity checks
"Your accuracy is 94%. What's your sensitivity?"	§11.1 — we never report accuracy; here is sens/spec with CI
"Did you tune on the test set?"	§10.4 — sealed set, frozen operating point with a recorded date
"Can you detect neovascularisation?"	§6.6 — no pixel ground truth in the named datasets; here is what we do instead

Question	Answer
"What happens when the model is wrong?"	\$4.2, \$8.6 — three-way output with escalation on disagreement
"How many graders does a district need?"	\$9.5 E1 — from the SimEvents sweep

14. Environment Setup

MATLAB has never been installed on the team's machine. Until `gpuDevice` returns the GPU, nothing else in this document can start. Treat this as the critical path.

14.1 Licensing

The \$99 MATLAB and Simulink Student Suite **does not include** Deep Learning Toolbox, Computer Vision Toolbox, Medical Imaging Toolbox or SimEvents **VERIFIED**. The team has no institutional Campus-Wide License. There is therefore **no default legal path** to the required toolchain, and this must be solved before any other work begins.

#	Action	Detail
1	Start a 30-day trial	Immediate unblock. Covers the internal round comfortably.
2	Email MathWorks India	Cite SIH26038 and request participant licences. MathWorks has partnered with SIH since 2019 and provides mentorship and software support for its problem statements VERIFIED . Send this on day one so a reply arrives before the trial lapses.
3	Check institutional access	Even without a Campus-Wide License, some institutions have limited departmental licences. Ask the department, not just the lab.

PRODUCTS TO SELECT ON THE TRIAL

MATLAB · Simulink · Deep Learning Toolbox · Image Processing Toolbox · Computer Vision Toolbox · Medical Imaging Toolbox · SimEvents · Statistics and Machine Learning Toolbox · **Parallel Computing Toolbox**

Parallel Computing Toolbox is not named in the PS tool list but is what enables GPU training. Omit it and you will train on CPU without understanding why everything takes twenty times longer.

The trial clock is a real risk. A trial started on 21 August expires around 20 September. The SIH grand finale is considerably later. Action 2 is not optional — it is what carries you to the finale. Track the expiry date in the risk register.

14.2 Installing on Arch Linux

MathWorks validates Ubuntu, Debian, RHEL and SUSE. Other distributions with kernel ≥ 4.18 and glibc ≥ 2.28 will "likely successfully run" MATLAB with limited support **VERIFIED**. Arch clears both thresholds comfortably. This is a support risk, not a technical blocker. The ArchWiki MATLAB page is the practical reference.

Two breakages you should expect

1. Bundled library conflict. MATLAB ships its own `libstdc++` and `libfreetype`, which conflict with Arch's newer system versions and produce segmentation faults or corrupted fonts. Force the system libraries: **VERIFIED**

```
export LD_PRELOAD=/usr/lib/libstdc++.so.6:/usr/lib/libfreetype.so.6
```

2. Blank window under Wayland. MATLAB's Java-based interface renders as a blank grey window under Wayland compositors via XWayland unless AWT is told the window manager does not reparent: **VERIFIED**

```
export _JAVA_AWT_WM_NONREParentING=1
```


Put both in a launcher script rather than typing them each session. You cannot avoid the GUI: App Designer and SimEvents both require it.

14.3 Smoke tests, in order

Test 1 — CUDA works at all.

```
matlab -batch "d = gpuDevice; disp(d.Name); disp(d.ComputeCapability); \
A = gpuArray.rand(4096); tic; B = A*A; wait(d); toc"
```

Expect the RTX 3050 and compute capability 8.6.

Test 2 — deep learning add-ons resolve. This forces a support-package download and confirms the toolbox licence is live.

```
matlab -batch "net = imagePretrainedNetwork('resnet50'); disp('DLT ok')"
```

Test 3 — SimEvents is available.

```
matlab -batch "simevents; disp('SimEvents ok')"
```

If all three pass, the environment is ready and the agent-driven build loop through `matlab -batch` can begin.

14.4 Hardware budget

GPU	RTX 3050 laptop, 6 GB VRAM — the binding constraint on batch size
RAM / storage	16 GB / 1 TB SSD — adequate
Dataset scale	~6,000 images across the four named datasets. Fine-tuning at 448×448 is a matter of hours, not days.
Cloud GPU	Not on the critical path. Optional, for larger-scale pretraining only. Roughly ₹500–800 for a GPU-day. Note that a rented VM needs its own MATLAB activation, so check licence terms first.

15. Risk Register

ID	Risk	Lik.	Impact	Mitigation
R01	MATLAB does not install or GPU unavailable	Med	Critical	Run §14.3 smoke tests immediately. Known Arch fixes documented in §14.2. Blocks everything, so resolve first.
R02	Trial licence expires before the finale	High	Critical	Email MathWorks on day one (§14.1 action 2). Record the expiry date and set a reminder two weeks ahead.
R03	Messidor-2 access not granted in time	Med	High	Request on day one. Fallback: patient-disjoint IDRiD split as a weaker external set, labelled as such (§10.1).
R04	MA segmentation fails on very small training set	High	Med	Track A classical morphological detector (§6.4) is built first and does not require training. Report both tracks.
R05	Neovascularisation cannot be segmented — no ground truth	Certain	Med	Declared in §6.6 and §17.1. Mitigated by image-level PDR detection plus mandatory escalation of Level 4.
R06	External validation misses the PS sens/spec targets	Med	Med	Anticipated in §10.5. Reported and analysed rather than tuned away. Reframed as the validation-rigour contribution (§11.4).
R07	No clinician available for the reader study	Med	Med	Fallback in §12.5: present the protocol as designed-but-unrun, report automated proxies. Never fabricate ratings.
R08	Model silently collapses to majority class	Med	High	Mandatory per-class recall logging and a zero-recall unit test (§13.4). The countermeasure is cheap; the failure is invisible without it.
R09	Team cannot defend design choices at judging	Med	High	This document. Rithviksai owns §1–§3, §11, §17 and the rationale notes.
R10	6 GB VRAM insufficient at target resolution	Med	Med	Reduce batch size before reducing resolution. Gradient accumulation if needed. Resolution is worth more than batch size here (§3.2).
R11	Simulink component left to the last minute	Med	Med	Independent of all ML work. Start it on day one, in parallel (§9).
R12	Self-inflicted train/inference preprocessing mismatch	Med	High	Single shared preprocessing function called from both paths (§5.4). Enforced by unit test.
R13	Patient-level data leak between splits	Med	High	Patient-level splitting rule (§10.2), split file committed, unit test asserting no patient ID appears in two splits.
R14	Only 2 of 6 team members actively building	Certain	Med	Accepted. Scope is set accordingly. Assign the SimEvents model and data curation to less technical members — both are high-visibility and need no ML.

16. Schedule and Ownership

16.1 Fixed dates

21 Aug 2026	Licensing, GPU smoke test, dataset access requests
25 Aug 2026	College internal hackathon — working demo mandatory, SIH judging criteria
30 Aug 2026	SPOC team nomination deadline
15 Sept 2026	SIH portal idea submission deadline
Sept → finale	Full build: learned segmentation, sealed external validation, reader study, SimEvents sweeps

16.2 Phase 1 — to 25 August: breadth over depth

Five components at demo fidelity beats one at research fidelity, because the judging rubric follows the PS's five requirements. Deliberately deferred: sparse BagNet, trained lesion segmentation, sealed external validation, reader study. Those become the roadmap slide, and a roadmap that names specific methods is itself a scoring signal.

Day	Build	In parallel
21 Aug	Licensing; §14.3 smoke tests; download APTOS and IDRiD; request Messidor-2	Recruit remaining team members
22 Aug	C1 quality gate — classical CV, no training. Start ResNet-50 fine-tune on APTOS.	SimEvents model skeleton
23 Aug	gradCAM + temperature calibration. Classical morphological MA detection for the lesion overlay.	SimEvents parameterisation and first sweep
24 Aug	App Designer UI: image → quality verdict → grade → Grad-CAM → lesion overlay → referral slip. Integrate.	Slide deck
25 Aug	Rehearse. Both presenters able to answer every question in §13.4.	—

DEMO HONESTY RULE

If IDRiD ground-truth lesion masks are shown to illustrate what the evidence layer will look like once trained, **label them as ground truth, not as model output**. Judges catch this, and volunteering the distinction reads as rigour rather than weakness.

16.3 Phase 2 — 26 August to 15 September

- Trained lesion segmentation on IDRiD (Track B, §6.4–§6.5)
- Vessel segmentation on DRIVE
- Ordinal head, uncertainty estimation, deferral policy
- Evaluation harness complete, ablation configurations A1–A5 runnable
- SimEvents experiments E1–E6
- Reader study scheduled
- Portal submission drafted from §1, §2, §11, §17

16.4 Phase 3 — to the grand finale

- Sparse BagNet arm and the XAI comparison of §11.7
- Operating point frozen and recorded in §11.2
- **Sealed set opened once** (§10.4)
- Reader study executed and reported as a pilot
- Full ablation table and Pareto curve produced

16.5 Ownership

Workstream	Owner	Notes
Presentation and defence	Rithviksai	Must be present in every design conversation, not handed slides at the end
ML pipeline (C1–C4)	Build pair	Driven through Claude Code; harness-first discipline per §13.4
SimEvents model (C5)	Assignable to a non-ML member	No GPU, no data, no ML. High visibility. Start day one.
Data curation and provenance	Assignable to a non-ML member	Access requests, §10.3 provenance log, quality-tier labelling
Clinician liaison	Any member	One email, then scheduling. Owns §12.

17. Known Limitations and Honest Disclosures

This section exists because the problem statement's central complaint is that existing solutions lack validation rigour. A limitations section that names real constraints is a direct answer to that complaint. Present it; do not bury it.

17.1 Neovascularisation

No pixel-level NV annotations exist in any of the four named datasets. We detect proliferative DR at image level through the classifier and use vessel-morphology proxy features as supporting indicators only. Every predicted Level 4 is escalated to a human regardless of confidence. **We do not claim a validated NV detector.** (§6.6)

17.2 Single-field imaging

Our pipeline grades single fundus photographs. The historical clinical reference standard for DR grading is seven-field stereoscopic photography. Single-field screening is standard practice and is what portable cameras produce, but it sees less retina, and peripheral pathology can be missed. This limits the ceiling on achievable sensitivity independently of model quality.

ESTABLISHED

17.3 Label noise

APTOS grades are single-grader without adjudication. Inter-grader agreement on DR severity is imperfect even among specialists, so a portion of apparent model error is label disagreement rather than model failure. This particularly affects the Level 1 versus Level 2 boundary — which is unfortunately our referral threshold. Report 5-class confusion matrices so adjacent-grade confusions are visible rather than hidden inside an aggregate. **ESTABLISHED**

17.4 Not a medical device

This is a screening aid and a research prototype. It has not been through any regulatory process, has no CDSCO clearance, and must not be used for clinical decision-making. Every generated report carries this statement in its footer (§8.7). Say this in the presentation too — it signals that you understand the domain rather than diminishing the work.

17.5 Diabetic macular oedema not addressed

The PS scopes the task to the ICDR severity scale and does not require DME detection. Clinically, DME is a distinct and important cause of vision loss and is part of a complete referral decision. Our hard-exudate detection with fovea-distance measurement (§8.4) provides a partial proxy, but this is **not** a DME classifier. Named here so the scope boundary is explicit.

17.6 Reader study scale

Planned as a single-reader, ~20-case pilot. Reported as a pilot with confidence intervals, never as clinical validation. (§12.4)

17.7 Dataset representativeness

APTOS and IDRiD are Indian, which is favourable for our deployment target, but neither was collected under the field conditions the PS describes — handheld cameras, uncontrolled lighting, untrained operators. Our quality gate is designed for those conditions but is validated on data that largely does not contain them. Closing this gap requires field data we do not have.

17.8 Simulation is not deployment

The SimEvents model uses parameters partly measured from our own pipeline (rejection rate, inference latency, grader time) and partly assumed (arrival patterns, bandwidth, connectivity). It is a capacity-planning tool whose conclusions are conditional on those assumptions. State the assumptions alongside every result it produces.

18. References

18.1 Verified during document preparation

Each of the following was consulted directly while writing this document. Claims tagged **VERIFIED** in the text trace to these.

#	Source	Used for
V1	Deep Learning Visualization Methods — MathWorks documentation	MATLAB provides exactly three image-explanation functions: <code>gradCAM</code> , <code>occlusionSensitivity</code> , <code>imageLIME</code> (§4.4, §8.2)
V2	<code>gradCAM</code> — MathWorks function reference	Grad-CAM implementation and layer selection (§8.2, §8.3)
V3	<code>occlusionSensitivity</code> — MathWorks function reference	Occlusion sensitivity; regression support from R2024a (§8.2)
V4	Understand Network Predictions Using LIME — MathWorks	Superpixel-based LIME for images (§8.2)
V5	An inherently interpretable AI model improves screening speed and accuracy for early diabetic retinopathy — <i>PLOS Digital Health</i> , 2025	Sparse BagNet architecture; lesion localisation precision 0.960 vs 0.656 for Guided Backpropagation; reader study 0.611 / 0.758 / 0.786; 11.7 s vs 15.2 s (§1.3, §8, §11.5). The single most important reference in this document.
V6	Linux System Requirements — MathWorks	Validated distributions; kernel ≥ 4.18 and glibc ≥ 2.28 for others (§14.2)
V7	MATLAB for Students — MathWorks	Student Suite contents; DLT / CVT / Medical Imaging / SimEvents are separate purchases (§4.4, §14.1)
V8	SimEvents — MathWorks product page	Entity Generator, Queue, Server, Resource Pool / Acquirer / Releaser, statistics blocks (§9.2)
V9	MathWorks Student Lounge — SIH team write-ups	MathWorks has partnered with SIH since 2019; provides mentorship and software support (§14.1)
V10	MATLAB — ArchWiki, and Arch Linux Forums threads on the <code>libstdc++</code> conflict	Bundled library conflict and the <code>LD_PRELOAD</code> fix (§14.2)
V11	MATLAB command options — MathWorks	<code>matlab -batch</code> : no desktop, logs to stdout/stderr, non-zero exit on failure (§13.4, §14.3)
V12	<code>_JAVA_AWT_WM_NONREParenting</code> — Wayland/XWayland Java AWT issue reports	Blank-window fix for MATLAB under Wayland compositors (§14.2)
V13	Smart India Hackathon portal — PS 26038 detail page	Problem statement text, organisation, category, theme, named datasets and toolboxes (§2.1)
V14	SIH 2026 team eligibility rules	Six student members per team, at least one female member

18.2 To be verified before citation

The references below are believed correct and are standard in the field, but were **not** confirmed against primary sources during document preparation. Each is tagged **UNVERIFIED** where it appears in the text. **Assign these, check them, and record the outcome in Appendix A before anything from them reaches a slide or the SIH submission.**

#	Claim to verify	Likely source to check
U1	IDx-DR pivotal trial sensitivity 87.2% / specificity 90.7% (§3.5)	Abrahamoff et al., autonomous AI diagnostic system pivotal trial, <i>npj Digital Medicine</i> , 2018
U2	EyeArt pivotal sensitivity ~96% / specificity ~88% (§3.5)	Eyenuk pivotal trial publication / FDA 510(k) summary

#	Claim to verify	Likely source to check
U3	~101 million adults with diabetes in India (§2.4)	ICMR-INDIAB national study, <i>Lancet Diabetes & Endocrinology</i> , 2023
U4	Saliency-map sanity checks; which methods fail parameter randomisation (§11.7)	Adebayo et al., "Sanity Checks for Saliency Maps", NeurIPS 2018
U5	Deletion / insertion AUC as a saliency faithfulness metric (§11.7)	Petsiuk et al., "RISE: Randomized Input Sampling for Explanation", 2018
U6	BagNet limited-receptive-field architecture (§1.3, §8)	Brendel & Bethge, "Approximating CNNs with Bag-of-local-Features models...", ICLR 2019
U7	Field image-rejection rate in the Thailand DR deployment study (§5.1)	Beede et al., human-centred evaluation of a deployed DR system, CHI 2020
U8	Exact dataset sizes and split conventions: APTOS, IDRiD, DRIVE, Messidor-2 (§10.1)	Confirm from the actual downloads. Fastest and most reliable check on this list.
U9	Quadratic weighted kappa as the APTOS 2019 competition metric (§11.3)	Kaggle APTOS 2019 competition evaluation page
U10	AUPR as the IDRiD lesion segmentation benchmark metric (§6.4)	IDRiD challenge documentation
U11	Local colour normalisation preprocessing and its attribution (§7.2)	Kaggle Diabetic Retinopathy Detection 2015 winning solution write-ups. Describe the method; do not attribute it to a named individual without checking.
U12	Retinal models can predict non-pathological attributes from fundus images (§4.2, §8.6)	Poplin et al., cardiovascular risk factor prediction from retinal fundus photographs, <i>Nature Biomedical Engineering</i> , 2018

Appendix A — Unverified Claims Register

Print this page. Assign each row. Nothing tagged Unverified appears in the SIH submission until its row is signed off, or the claim is deleted.

Ref	Claim	Section	Assigned to	Checked	Outcome
U1	IDx-DR 87.2% / 90.7%	§3.5			
U2	EyeArt ~96% / ~88%	§3.5			
U3	India ~101M adults with diabetes	§2.4			
U4	Saliency sanity-check failures	§11.7			
U5	Deletion/insertion AUC	§11.7			
U6	BagNet architecture reference	§1.3			
U7	Thailand deployment rejection rate	§5.1			
U8	Dataset sizes and splits	§10.1			
U9	APTOS metric = QWK	§11.3			
U10	IDRiD metric = AUPR	§6.4			
U11	Colour normalisation attribution	§7.2			
U12	Non-pathological attribute prediction	§4.2			
—	DRIVE 20/20 split sizes	§6.3			
—	APTOS class distribution	§7.4			
—	SIH judging theme bucket for PS 26038	§2.1			

U8 and the APTOS class distribution are the highest priority: both are answered in minutes by inspecting the downloaded files, and both appear in the pitch.

Appendix B — Day One Checklist

Everything on this page is blocking, and none of it depends on any design decision. Work top to bottom.

□	Action	Owner	Why it is blocking
□	Start the 30-day MathWorks trial with all nine products from §14.1, including Parallel Computing Toolbox		Nothing else can begin
□	Email MathWorks India citing SIH26038, requesting participant licences		The trial expires around 20 Sept; the finale is later (R02)
□	Install MATLAB; apply both §14.2 fixes in a launcher script		Known Arch and Wayland breakages
□	Run §14.3 smoke tests 1, 2 and 3		Confirms GPU, deep learning toolbox and SimEvents all work (R01)
□	Download APTOS 2019 and IDRiD		Training data
□	Request Messidor-2 access (licence agreement, can take days)		Sealed external test set (R03)
□	Download DRIVE		Vessel segmentation
□	Create <code>data/PROVENANCE.md</code> ; record sizes, dates, licences, hashes		§10.3; also answers Appendix A rows U8 and the class distribution
□	Confirm the SIH judging theme bucket with the SPOC		Portal lists this MedTech PS under Clean & Green Technology (§2.1)
□	Open the SimEvents workstream — independent of all ML work		R11; highest return per hour in the PS
□	Email an ophthalmology department to request one reader for two hours		§12; long lead time, one email
□	Write <code>eval/harness.m</code> before the first model		R08 — the most important engineering practice in this document

THE THREE THINGS THAT DECIDE THIS PROJECT

1. **Get MATLAB running today.** Everything is downstream of a working `gpuDevice`.
2. **Build the evaluation harness before the first model.** A model that silently collapses to the majority class raises no error, and an AI-driven build loop will report it as success.
3. **Never report a number you did not measure.** The problem statement's own complaint is that existing solutions lack validation rigour. Being the team whose numbers are real, whose external result is reported honestly, and whose limitations are stated openly is not a consolation strategy — on this problem statement, it is the winning one.